

23/07/2026

ASSESSMENT TOOL-IAnalytical Problem Solvings

① Given,

Total tosses ( $n$ ) = 10Number of Heads ( $k$ ) = 7

Number of Tails = 3

w.k.T

$$L(P) = P^k (1-P)^{n-k} \rightarrow \text{Likelihood Function}$$

$$= P^7 (1-P)^3$$

$$\ln L(P) = 7 \ln(P) + 3 \ln(1-P)$$

$$\frac{d}{dP} \ln L(P) = \frac{7}{P} - \frac{3}{1-P} = 0$$

$$\Rightarrow 7(1-P) = 3P$$

$$\Rightarrow 7 - 7P = 3P$$

$$\Rightarrow 10P = 7$$

$$\Rightarrow P = \frac{10}{7}$$

$$= 0.7 \text{ (or) } 70\%$$

②

Given,

Total Students ( $n$ ) = 50;Students passed ( $k$ ) = 42

The maximum Likelihood Estimator for a Bernoulli trial parameter  $p$  is the sample mean

$$P = \frac{k}{n}$$

$$= \frac{42}{50}$$

$$= 0.84$$

$$\therefore \hat{p} = 84\%$$

③ Given,

Total Sample size ( $n$ ) = 100

Defective bulbs ( $k$ ) = 8

By using the MLE formula for proportion:-

$$\begin{aligned}\hat{p} &= \frac{k}{n} = \frac{8}{100} \\ &= 0.08\end{aligned}$$

$$\therefore \hat{p} = 8\%$$

④ Given,

Total rolls ( $n$ ) = 60

Occurrences of 6 ( $k$ ) = 18

By using the MLE formula:-

$$\begin{aligned}\hat{p} &= \frac{k}{n} \\ &= \frac{18}{60} \\ &= 0.3\end{aligned}$$

$$\therefore \hat{p} = 30\%$$

⑤ Given,

Input ( $x$ ) = 4

weight ( $w$ ) = 2

Actual Output ( $y_{\text{actual}}$ ) = 10

Prediction model:  $y = w \times x$

Step 1:- Finding the Predicted Output:-

$$\hat{y} = w \times x$$

$$= 2 \times 4$$

$$= 8$$

Step-2:- calculate error:-  
mm mm mm mm mm

$$\text{Error} = Y_{\text{actual}} - \hat{y}$$

$$= 10 - 8$$

$$= 2$$

Step-3:- update weight:-  
mm mm mm mm

Given,

$$\text{Gradient} (\nabla w) = 4$$

$$\text{learning rate} (\eta) = 0.01$$

By using the gradient Descent weight update rule

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot \text{gradient}$$

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04$$

$$= 1.96_{,,}$$